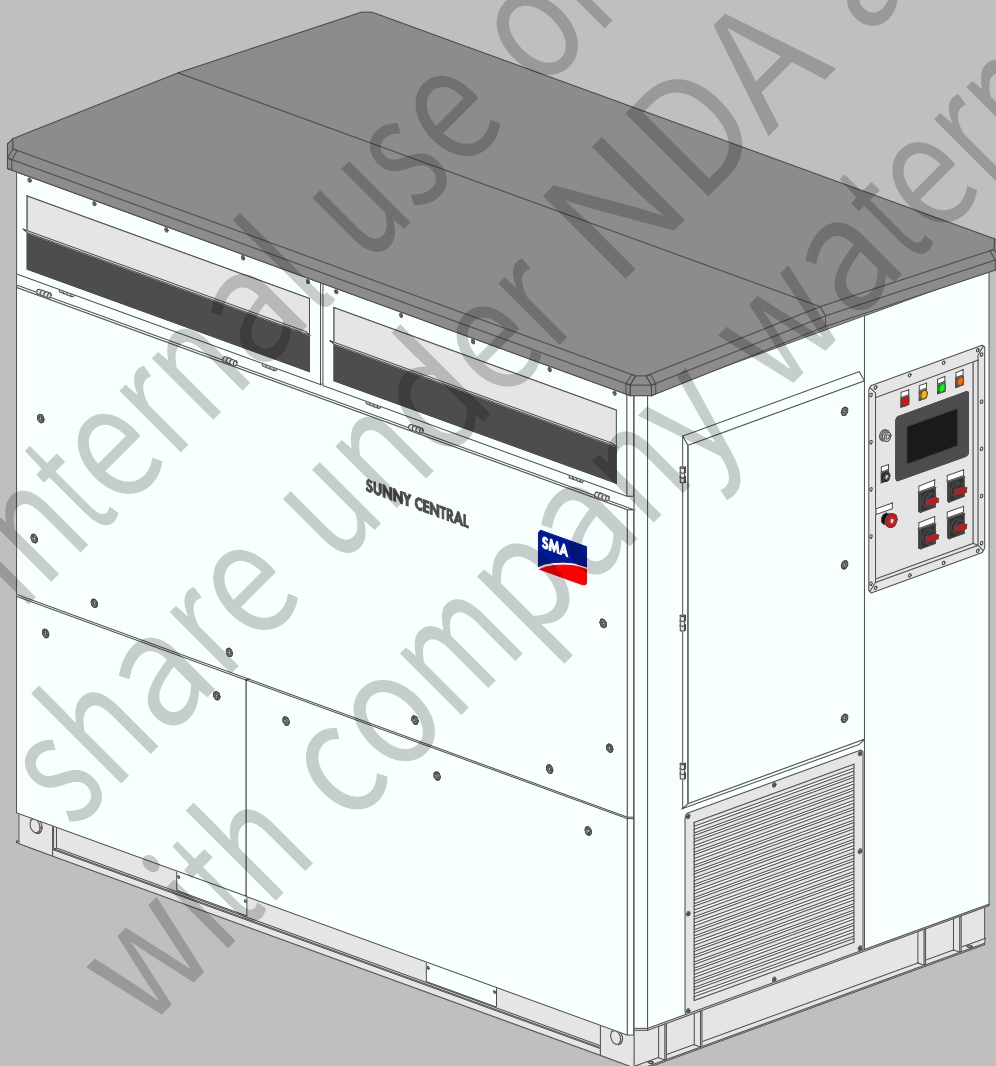




Technical Information
**Modbus® Interface for
SUNNY CENTRAL**



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1 Information on this Document

1.1 Validity

This document is valid for the following device types from firmware version 10.03.xx.R:

- Sunny Central 2660 UP (SC 2660 UP)
- Sunny Central 2800 UP (SC 2800 UP)
- Sunny Central 2930 UP (SC 2930 UP)
- Sunny Central 3060 UP (SC 3060 UP)
- Sunny Central 4000 UP (SC-4000UP)
- Sunny Central 4200 UP (SC-4200UP)
- Sunny Central 4400 UP (SC-4400UP)
- Sunny Central 4600 UP (SC-4600UP)
- Sunny Central xxxx UP (2 Stack)
- Sunny Central xxxx UP (3 Stack)
- Sunny Central 2660 UP-US (SC 2660 UP-US)
- Sunny Central 2800 UP-US (SC 2800 UP-US)
- Sunny Central 2930 UP-US (SC 2930 UP-US)
- Sunny Central 3060 UP-US (SC 3060 UP-US)
- Sunny Central 4000 UP-US (SC-4000UP-US)
- Sunny Central 4200 UP-US (SC-4200UP-US)
- Sunny Central 4400 UP-US (SC-4400UP-US)
- Sunny Central 4600 UP-US (SC-4600UP-US)
- Sunny Central xxxx UP-US (2 Stack)
- Sunny Central xxxx UP-US (3 Stack)

It describes the variant of the communication protocol "Modbus® Application Protocol" implemented by SMA Solar Technology AG and the corresponding parameters, measured values and data exchange formats.

This document does not contain any information on software which can communicate with the Modbus interface (see the software manufacturer's manual).

1.2 Target Group

This document is intended for qualified persons. The activities described in this document are to be performed by qualified persons only. Qualified persons must have the following skills:

- Knowledge of IP-based network protocols
- Knowledge of installation and configuration of IT systems
- Knowledge of how SMA inverters work and are operated
- Training in how to deal with dangers and risks associated with operating electrical devices and systems
- Knowledge of and compliance with this document

1.3 Additional Information

SMA documents

Additional information is available at www.SMA-Solar.com:

Additional Documents

Document title	Source
Service Name and Transport Protocol Port Number Registry	http://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xml
Modbus Application Protocol Specification	http://www.modbus.org/specs.php

1.4 Symbols

Symbol	Explanation
	Information that is important for a specific topic or goal, but is not safety-relevant.
	Indicates a requirement for meeting a specific goal.
	Desired result.
	A problem that might occur.

1.5 Typographies

Typography	Use	Example
bold	File names Elements to be selected Elements on a user interface Elements to be entered Parameter	Select Settings . Read off the configured port in the field Port . Enter the value 502 in the field Port
>	Connects several elements to be selected	Select Settings > Port .
[Button/Key]	Button or key to be selected or pressed	Select [Next].
[Source]	Data channel name	Reading of the system time (UTC) [SerTm]

1.6 Nomenclature

Complete designation	Designation in this document
SMA Modbus Master Profile	Modbus master profile
Sunny Central	Inverter

1.7 Terms and Abbreviations

Term/Abbreviation	Designation	Explanation
Attribute value	-	Changeable content of an XML attribute, e.g., attribute="My content".

Term/Abbreviation	Designation	Explanation
CT	Current measuring unit	Hardware for string-current monitoring
DWORD	-	Data with a width of 32 bits, as per IEC 61131-3
Device ID	-	Numerical value that identifies a certain SMA device type, e.g., 102 = SMA Power Plant Controller.
GFDI	Ground-Fault Detection and Interruption	Detection of the grounding error and subsequent interruption of the electric circuit.
Hex	-	Hexadecimal number
IP	Internet Protocol	Network protocol for connections via the Internet.
MPP	Maximum Power Point	Maximum power point
NaN	Not a Number	No valid value is available
PMAX	Active power, maximum value	The device can generate active power up to this limit.
Source register	-	Modbus register that is assigned to a different Modbus address (destination register).
RO	Read Only	Value can only be read
RW	Read/Write	Value can be read and written
SCADA	Supervisory Control and Data Acquisition	Control center software. In this document, SCADA is used as an example of a Modbus master system.
UTC	Coordinated Universal Time	Coordinated Universal Time
VPN	Virtual Private Network	VPN is used to establish a connection from one private computer network to another. The point of connection is established on both sides by a VPN gateway. In each case, the VPN gateway may be a computer or a router. The connection itself is described as a VPN tunnel. A secured data stream runs via the tunnel from one network to the other, providing VPN nodes from one network with access to services and devices of the other network.
WO	Write-only value	Value can only be written
WORD	-	Data with a width of 16 bits, as per IEC 61131-3
Destination register	-	Modbus register to which a different Modbus address (source register) is assigned. The destination register is used instead of the source register for further data processing.

2 Security

2.1 Intended Use

The Modbus interface of the inverter is designed for industrial use and has the following tasks:

- Remote control of the grid management services of a PV system.
- Remote-controlled querying of the measured values of a PV system.
- Remote-controlled changing of the parameters of a PV system.

The Modbus interface can be accessed via the LAN port of the inverter and it can be used via the protocol Modbus TCP and by the protocol Modbus UDP.

The enclosed documentation is an integral part of this product. You must read and observe the documentation and keep it in a convenient place for future reference.

2.2 Information on Data Security

Data Security in Ethernet Networks

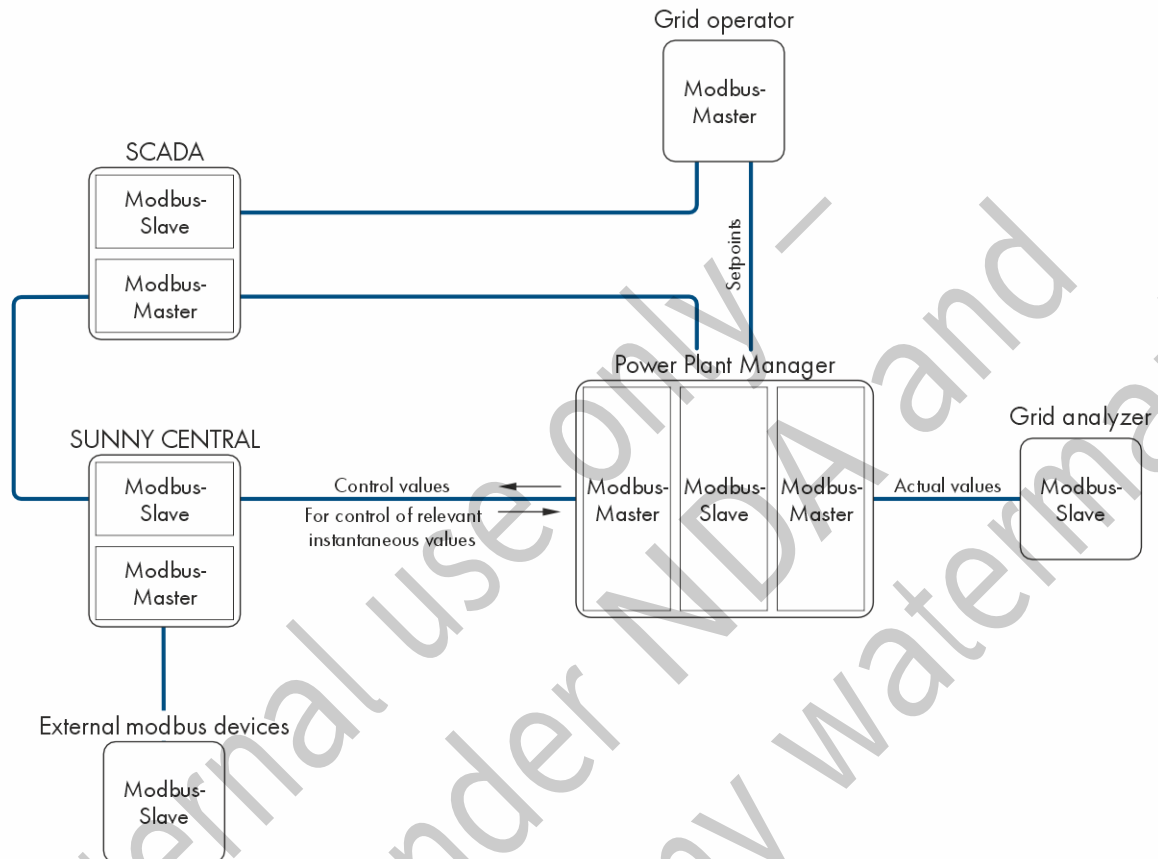
You can connect the inverter to the Internet. When connecting to the Internet, there is a risk that unauthorized users can access and manipulate the data of your system.

- Take appropriate protective measures, for example:
 - Set up a firewall
 - Close unnecessary network ports
 - Only enable remote access via VPN tunnel
 - Do not set up port forwarding at the Modbus port in use

3 Product Description

3.1 Modbus Protocol

The Modbus Application Protocol is an industrial communication protocol that is currently used in the solar sector mainly for system communication in PV power plants. The Modbus protocol is based on a master/slave architecture.



The Modbus protocol has been developed for reading data from or writing data to clearly defined data areas. The Modbus specification does not prescribe what data is within which data area. The data areas must be defined device-specifically in Modbus profiles. With knowledge of the device-specific Modbus profile, a Modbus master (e.g., a SCADA system) can access the data of a Modbus slave (e.g. communication interface of the inverter).

The special Modbus profile for SMA devices is the SMA Modbus profile.

3.2 SMA Modbus Profile

The SMA Modbus profile contains definitions for selected SMA devices. A reduction of the available data on SMA devices was carried out for the definition and this was then assigned to the corresponding Modbus registers. The SMA Modbus profile contains, for example, the total and daily energy, current power, voltage and current levels. The assignment between SMA device data and Modbus addresses is split into ranges in the SMA Modbus profile and these can be addressed via unit IDs (see Section 3.6 "Addressing and Data Transmission in the Modbus Protocol", page 11).

To enable access to the data of an SMA device, a special gateway is required, which is provided via the communication interface of the inverter.

3.3 User-Defined Modbus Profile

The user-defined Modbus profile enables you to reassign Modbus addresses of the SMA Modbus profile.

One advantage of reassigning Modbus addresses can be that the measured values and parameters which are relevant for a specific purpose, can be applied to consecutive Modbus addresses. These addresses can then be read or set in a single data block.

3.4 SMA Modbus Master Profile

The SMA Modbus master profile is an extension of the Modbus interface, which is used to access Modbus-capable slave devices of other providers. Each external Modbus device has a separate profile (*.mmp) and is managed as a separate device in the inverter. The external devices can be accessed under the unit IDs = 16 to 31 via the Modbus slave in the inverter.

3.5 System Topology

The SMA Modbus profile was developed for a hierarchical system structure. In this structure the inverter has a communication interface that is equipped with a Modbus TCP/IP and Modbus UDP interface. All other Modbus-capable devices are subordinate to the inverter.

From the perspective of the Modbus protocol, the inverter represents a Modbus slave that provides a gateway to the Modbus-capable devices. These devices can only be addressed via the gateway of the inverter by unit ID.

3.6 Addressing and Data Transmission in the Modbus Protocol

3.6.1 Unit ID's

The Unit ID is a superordinate addressing type in the Modbus protocol. The SMA Modbus protocol has 247 Unit IDs, of which 245 can be assigned to individual devices. If a unit ID is assigned to a device, then the parameters and measured values of this device can be accessed.

The following table shows an overview of the unit IDs in the SMA Modbus profile: Individual unit IDs and unit-ID ranges are reserved and are not used with automatic system detection. With manual detection, however, these can be used:

Unit ID	Explanation
2	This unit ID is reserved for the system parameters.
3	This unit ID is reserved for the inverter.
4 to 15	Reserved
16 to 31	External devices. The last character of the IP address of the external device corresponds to the assigned unit ID.
32	Left DC module for zone monitoring
33	Right DC module for zone monitoring
34	Center DC module for zone monitoring This DC module is only available when using the "DC coupling with 36 single-pole fused inputs" option.
34 to 101	Reserved
102	SMA Power Plant Controller or SMA Power Plant Manager
103 to 119	Reserved
120 to 169	SMA String-Monitor (SSM-U-xxxx)
170 to 247	Reserved
255	Devices that are assigned to this unit ID cannot be addressed. You must assign unit IDs from the range 3 to 247 to these devices.

3.6.2 Assignment of the Modbus Register to Unit IDs

The assignment of the parameters and measured values of the SMA devices to Modbus register addresses is achieved using assignment tables and is also shown in this document (see Section 5 "SMA Modbus Profile—Assignment Tables", page 20).

In the assignment table "PV System Parameters (unit ID = 2)", parameters and measured values of the inverter and of the PV system are saved.

In the assignment table "Inverter (unit ID = 3)", the parameters and measured values used in the inverter are saved.

In the assignment tables "External devices (unit ID = 16 to 31)", the channels used for the various external devices are saved.

3.6.3 File Formats Used

The Modbus protocols are saved in XML format. For easier assignment, the profiles of the various devices have different file extensions.

Expansion	Explanation
*.kmp	Modbus profile of the inverter
*.vdp	User-Defined Modbus Profile
*.mmp	Modbus profile of external devices

3.6.4 Modbus Register Address, Register Width and Data Block

A Modbus register is 16 bits wide. For wider data types, connected Modbus registers are used and considered as data blocks. The number of connected Modbus registers is indicated in the assignment tables. The address of the first Modbus register in the data block is the start address of the data block. In addition, larger data blocks can be formed.

3.6.5 Address Range for the Modbus Register

For addressing Modbus registers, the address range 0 to 0xFFFF is available with 65536 addresses.

3.6.6 Data transmission

In accordance with the Modbus specification, only a specific volume of data can be transported in a single data transmission in a simple protocol data unit (PDU). The data also contains function-dependent parameters such as the function code, start address or number of Modbus registers to be transmitted. The amount of data depends on the Modbus command used and has to be taken into account during data transmission. You can find the number of possible Modbus registers per command in Section 3.8.

With data storage in the Motorola format "Big Endian", data transmission begins with the high byte and then the low byte of the Modbus register.

3.7 Address Types

The individual Modbus registers can be addressed with different data.

Address type	Explanation	Modbus function code
Coil	One bit, which can be read, is set, e.g., for digital signals.	FC 01 (0x01) – Read coils FC 05 (0x05) – Write single coil FC 15 (0x0F) – Write multiple coils
Discrete	One bit, which can be read and written, is set, e.g., for digital signals.	FC 02 (0x02) – Read discrete inputs
Input	The addressed registers contain measured values and can only be read (RO).	FC 04 (0x04) – Read input registers
Holding	The addressed registers contain parameters and can be read and written (RW).	FC 03 (0x03) – Read holding registers FC 06 (0x06) – Write single holding register FC 16 (0x10) – Write multiple holding registers

3.8 Reading and Writing Data in the Modbus Protocol

The Modbus interface can be used via the protocol Modbus TCP and by the protocol Modbus UDP. Using Modbus TCP enables read and write access (RW) and using Modbus UDP enables only write access (WO) to the Modbus register.

The following Modbus commands are supported by the implemented Modbus interface:

Modbus command	Modbus function code	Data volume (number of registers) ¹
Read Coils ²	FC 01 (0x01)	1 to 2000
Read Discrete Input	FC 02 (0x02)	1 to 2000
Read Holding Registers	FC 03 (0x03)	1 to 123
Read Input Registers	FC 04 (0x04)	1 to 125
Write Single Coil	FC 05 (0x05)	1
Write Single Holding Register	FC 06 (0x06)	1
Write Multiple Coil	FC 15 (0x0F)	1 to 1968
Write Multiple Registers	FC 16 (0x10)	1 to 123

Reading or writing a single Modbus register

If a Modbus register is accessed that is not contained in the Modbus profile, or if a Modbus command is incorrect, a Modbus exception is generated. Modbus exceptions are also generated when write access occurs on a read-only Modbus register or read access occurs on a write-only Modbus register.

Reading or writing of data blocks

To prevent inconsistencies, data blocks of associated registers or register ranges must be read or written consecutively.

Writing multiple Modbus registers as a data block

If multiple registers are written in the data block (Modbus command 0x10) and an error occurs when writing, the next register in the data block is used to proceed. If some data is dependent on other data, or if some data is mutually exclusive, the data is only processed if the entire data block is valid. Otherwise the entire data block is discarded. In the event of an error, a Modbus exception will be generated.

Reading multiple Modbus registers as a data block

If a data block is read and if at least one register defined in the Modbus profile can be determined in its data range, an answer is returned. If none of the Modbus registers are defined in the data range of a data block in the Modbus profile, the query is invalid, and a Modbus exception is generated.

¹ Number of Modbus registers transferable as a data block per command (16 bit)

² Not available for the SMA Modbus master profile.

Modbus exceptions

For Modbus exceptions, see "Modbus Application Protocol Specification" at <http://www.modbus.org/specs.php>.

3.9 SMA Data Types

3.9.1 Data types

The following table shows the data types used in the SMA Modbus profile. The SMA data types are listed in the assignment tables in the **Type** column. They describe the data widths of the assigned values:

Type	Description
U16	A word (16-bit/WORD) in the local processor format
S16	A signed word (16-bit/WORD) in the local processor format
U32	A double word (32 bit/DWORD) in the local processor format
S32	A signed double word (32 bit/DWORD) in the local processor format

3.9.2 16-Bit Integer Values

16-bit integers are stored in a Modbus register.

Modbus register	1	
Byte	0	1
bits	8 ... 15	0 ... 7

3.9.3 32-Bit Integer Values

32-bit integers are stored in two Modbus registers.

Modbus register	1		2	
Byte	0	1	2	3
bits	24 ... 31	16 ... 23	8 ... 15	0 ... 7

Example

DcMs.TotAmp: UnitId: 3 Register: 8 Data type: S32 Fix0 Example value: -269 A

Modbus register	8	9
Value	65535	65267

InvMs.DclVol.Stk1: UnitId 3 Register 12 Data type S32 Fix1 Example value: 842.1 V

Modbus register	12	13
Value	0	8421

3.10 SMA Data Formats

The following SMA data formats describe how SMA data is to be interpreted. The data formats are used, for example, for the display of data or for its further processing. The SMA data formats are listed in the **Format** column of the assignment tables.

Format	Explanation
Duration	Duration Time in seconds, in minutes or in hours, depending on the Modbus register.
DT	Date/Time Date/time, in accordance with country setting. Transmission as UTC (seconds since 1970-01-01).
FIX0	Factor 1 Decimal number, commercially rounded, no decimal place.
FIX1 to FIX8	Scaling factor to define the decimal places (up to 8) Decimal number, commercially rounded, number of decimal places in accordance with the number (FIX2 corresponds to a decimal number with two decimal places)
ENUM	Coded numerical values. Display of settings in abbreviations, e.g., for operating states or selected procedures for setpoint. Here it is only possible to select one numerical value in each case. The breakdown of the possible codes can be found directly under the designation of the Modbus register in the SMA Modbus profile – assignment tables.
ENUM_FLAG	Coded numerical values. Display of settings in abbreviations, e.g., for operating states or selected procedures for setpoint. Here it is possible to select multiple numerical values.
VERSION	Firmware version (see "Firmware version extract" below).
TIME_DDMMYY	Display as date, e.g., "020913" for September 2, 2013
TIME_YYYYMMDD	Display as date, e.g., "20130902" for September 2, 2013
TIME_HHMMSS	Display as time, e.g., "134725" for 13:47:25
TIME_YYYYMMDDHHMMSS	Display as date and time, e.g., "20130902134725" for September 2, 2013, 13:47:25
BITMASK	Assignment of meaning of the individual bits of the register
PLAINSTRING	Display as text
ARRAY	Display as text
IPV4	Display as IP V4 address, e.g., "192.168.0.5"
IPV4_ADDR_LIST	Display as list of IP V4 addresses
IPV4_ADDR_MASK	Display as IP V4 address and net mask
IPV6	Display as IP V6 address, e.g., "2001:0db8:85a3:08d3:1319:8a2e:0370:7344"
MAC	Display as MAC address, e.g., "12-34-56-78-9A-BC"
DURATION	Display as duration, e.g., days h:min:sec or h:min:sec

Format	Explanation
ACTION	Display as visual button
ACTION_CONFIRMED	Display as visual button with option to confirm
BOOL	Display as checkbox or any other boolean display

Abstract firmware version, format VERSION

Four values are extracted from the delivered DWORD. The values **Major** and **Minor** are contained BCD-coded in bytes 1 and 2. Byte 3 contains the **Build** value (not BCD-coded).

Byte 4 contains the **Release Type** value according to the following table:

Release type	Release-type coding	Explanation
3	B	Beta release
4	R	Release
5	S	Special release
> 5	As number	No special interpretation

Example

Firmware version of the product:

1.5.10.R

Values from DWORD:

Major: 1, Minor: 5, Build: 10, Release type: 4
(Hex: 0x1 0x5 0xA 0x4)

3.11 Content-Based Grouping of the Registers

The Modbus addresses described below are differentiated into three basic groups according to their function:

- Setpoints (S)
- Parameter (P)
- Measured values (M)

For optimum system control, a distinction must be made between these groups with regard to their use.

Setpoints

Setpoints encompass the requirements for the current operating point. Here, the signals for active and reactive power (**Wspt** and **VarSpt**) are the most important registers because they are also used by the inverter as a sign of life to monitor the communication connection. The processing of these signals is optimized for frequent transmission at high speed. Transmission can therefore be carried out, for example, every 100 ms. However, the cycle time should not be less than 50 ms to avoid an overload of the signal processing. To enable communication monitoring, these setpoints for active and reactive power must be sent independently of the current operating state.

Parameter

Parameters are generally provided as holding registers and can be read to determine the settings of the inverter. If you have write access, the registers can be written to change the settings of the inverter.

Every time these values are written, the settings database stored in the non-volatile memory is updated. These registers should only be written if the current setting needs to be adjusted.

Measured Values

Measured values are provided as input registers and can only be read. They contain information about the current inverter state and provide currently measured values of the inverter. The values are updated in every 100 ms cycle. The polling rate should preferably correspond to a multiple of this internal refresh rate.

Not all registers have to be used for correct communication.

Which registers are actually required should be decided depending on the application.

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4 Configuration

4.1 Detection of SMA Modbus Devices and Allocation of Unit IDs

With a new PV power plant, the Modbus is activated on commissioning and the Modbus-capable devices are detected on commissioning. The Modbus-capable SMA devices log in to the inverter independently and the inverter assigns them a unit ID.

If new SMA devices are added to the system or if devices must be replaced, the new devices are also detected automatically.

4.2 Importing the Modbus Profile File of External Devices

External Modbus devices can be connected to the inverter. In order to be able to access these devices via the Modbus protocol also, the external Modbus devices must be detected in the system.

Procedure:

1. Open Modbus profile of the external device in a UTF-8-capable editor (e.g., Notepad++).
2. In the Modbus profile, enter the IP address under which the device can be accessed.
3. Save change. Give the file a unique name. If multiple files of the same name are used, during an import the existing file will be overwritten in each case by the imported file.
4. Log in to the user interface of the inverter as installer (see the inverter user manual).
5. Import the changed Modbus profile (see inverter user manual). Here, select **Modbus Profiles** from the list of available types. A unit ID contained in the profile is assigned to the device automatically.

4.3 Exporting the Profile File

1. Log in to the user interface of the inverter as installer (see the inverter user manual).
2. In the main navigation, select **Export** (see inverter user manual).
3. Select **Modbus profiles** from the list of available types.
4. Select the Modbus profile to be exported.
5. If the profile file is to be saved to an external storage medium that is present in the inverter, select the **[Copy file to external device]** button.
If the profile file is to be saved on the computer, select the **[Download]** button.
6. Choose a storage location for the profile file and save it.

4.4 Deleting the Profile File

1. Log in to the user interface of the inverter as installer (see the inverter user manual).
2. In the main navigation, select **Export** (see inverter user manual).
3. Select **Modbus profiles** from the list of available types.
4. Select the Modbus profile to be deleted.
5. Select the button [**Delete files**].

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5 SMA Modbus Profile—Assignment Tables

5.1 Information on the Assignment Tables

The following subsections are sorted by unit ID. Each contains a table of the Modbus addresses which can be accessed using this unit ID. The tables present the following information:

Information	Explanation
Address	Decimal Modbus address (see Section 3.6 and onwards)
Channel name	Name of the channel
Explanation	Short description of the data channel of the Modbus register
Unique ID	Assignment to the channels on the user interface of the inverter
Data type	Data type, e.g., U32 = 32 bits without prefix (see Section 3.6)
Data format	Data format of the saved value, e.g., DT = date, FIXn = output with n decimal places, TEMP = output as temperature (see Section 0)
Access	Access type for Modbus TCP (see Section 3.8 "Reading and Writing Data in the Modbus Protocol", page 13): RO: Read only RW: Read and write If an access type is not supported, a Modbus exception will be generated.
Scaling	Specification of scaling factor with which the data must be converted
Offset	Specification of offset with which the data must be converted.
Unit	Specification of the unit in which the data is specified

5.2 System Parameters (Unit ID = 2)

The PV system parameters represent measured values and parameters of the inverter and also PV system devices that are connected via the Modbus protocol. Parameters such as time settings are transferred by the inverter to the devices of the PV system and there, depending on the device type, processed further. Measured values such as energy meter values are queried by the devices and made available as accumulated values.

5.2.1 Holding Register

In the following table, you can find the measured values supported by the inverter, which you can access under unit ID = 2 and function code 0x03, 0x06 and 0x10 (see Section 3.6.1 "Unit ID's", page 11).

Parameter

Address	Description / number code	Data type	Format	Access
1104	Setting of the system time (UTC)	U32	DT	RW
1106	Setting of the time zone. For possible values, see Section 9.5 "Number Codes of the Time Zones", page 58	U32	ENUM	RW

Setpoints

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Data format	Unit
40018	FstStop	Disconnection of the inverter	-	U32	1	ENUM	-
40022	VArSpt	Reactive power setpoint Q, in % of the maximum reactive power (referred to VArExlSpt.RefVal ; the reference value can be defined via the parameter VArExlSpt.RefMod) of the inverter. Value range: -100.00% to < 0% = underexcited 0% = no reactive power 0% to +100.00% = overexcited	-	S16	100	FIX2	%
40023	WSpt	Active power setpoint P, in % of the maximum active power (referred to WExlSpt.RefVal ; the reference value can be defined via the parameter WExlSpt.RefMod) of the inverter. Value range: 0% = No active power < 0% to +100.00% = generator or discharge mode -100.00 % to < 0 % = load or charge mode	-	S16	100	FIX2	%
40024	VArModCfg.P FCtlComCfg.PF	Displacement power factor cos φ 0.0000 to 0.9999	-	U16	10000	FIX4	-

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Data format	Unit
40025	VArModCfg.P FCtlComCfg.PF Ext	Excitation type of $\cos \varphi$	-	U32	1	ENUM	-
41261	HzNomSpt	Nominal frequency for frequency-dependent control of active power / frequency setpoint	-	U32	1000	FIX3	Hz
41263	VolNomSpt	Nominal AC voltage for voltage-dependent reactive power control / AC voltage setpoint	-	U16	10000	FIX4	p.u.
44039	WSptMax	Higher priority limitation of the active power setpoint WSpt in % of the maximum active power of the inverter (referred to WExlSpt.RefVal ; the reference value can be defined via the parameter WExlSpt.RefMod). Value range: 0% = No active power < 0% to +100.00% = generator or discharge mode Use via communication must be enabled by GriMng.WMaxMod .	-	S32	100	FIX2	%
44041	WSptMin	Lower limitation of the active power setpoint WSpt in % of the maximum active power of the inverter (referred to WExlSpt.RefVal ; the reference value can be defined via the parameter WExlSpt.RefMod). Value range: 0% = No active power < 0% to +100.00% = generator or discharge mode -100.00 % to < 0 % = load or charge mode Use via communication must be enabled by GriMng.WMinMod .	-	S32	100	FIX2	%

5.2.2 Input Register

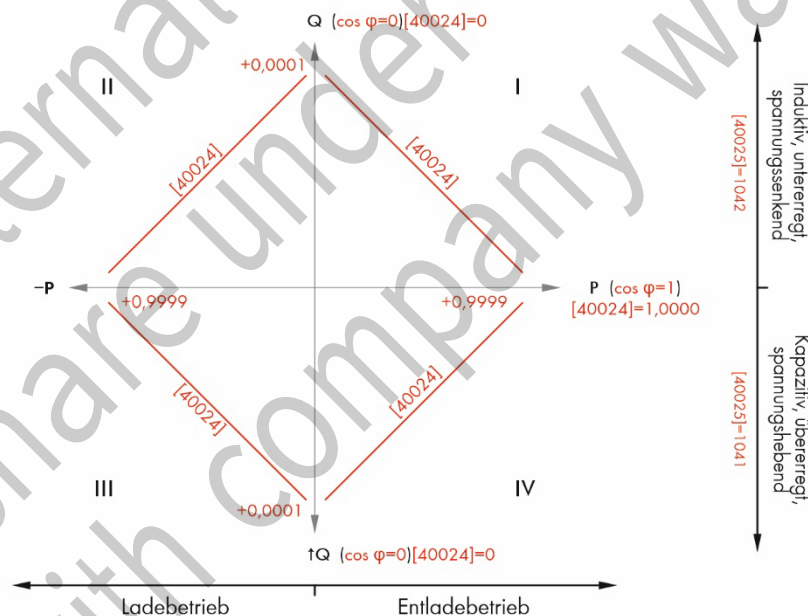
In the following table, you can find the parameters supported by the inverter, which you can access under unit ID = 2 and function code 0x03, 0x06 and 0x10 (see Section 3.6.1 "Unit ID's", page 11).

Measured Values

Address	Description / number code	Data type	Format	Access
0	Version number of the SMA Modbus profile	U32	FIX0	RO
2	Device ID of the communication interface	U32	FIX0	RO
6	Modbus data change: Meter value is increased on each access.	U32	FIX0	RO
300	Reading of the system time (UTC)	U32	DT	RO
700	Upd.PkgVer	U32	VERSION	RO
709	Upd.PkgVer.Dec	U32	FIX0	RO

5.2.3 Power Control with $\cos \varphi$ and Excitation Type

The power control by means of $\cos \varphi$ [40024] and the excitation type [40025] are shown in a coordinate system in the following graphic (the square brackets contain the address of the register to be set):



The power can be controlled either via the desired reactive power value [40022] or via the displacement power factor $\cos \varphi$ and the type of excitation. In this case, the power control with $\cos \varphi$ [40024] and the excitation type [40025] is active only if the parameter **Gri.Mng.VarMod** is set to **PFctlCom**.

5.3 Inverter Parameter (Unit ID = 3)

5.3.1 Holding Register

In the following table, you can find the parameters supported by the inverter, which you can access under unit ID = 3 and function codes 0x03, 0x06 and 0x10 (see Section 3.6.1, "Unit ID's" page 11).

Channels marked with * have been prepared for the SunSpec profile.

Parameter

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
0	InvOpMod	Operating state of the inverter, setpoint	329	S32	1	0	ENUM	-
2	RemRdy	Put inverter into operation or standby	331	S32	1	0	ENUM	-
4	GriMng.VArMod	Grid management service: Selection of procedure for setpoint for reactive power	6080	S32	1	0	ENUM	-
6	GriMng.WMod	Utility grid system services: Selection of procedure for setpoint for active power	6078	S32	1	0	ENUM	-
8	ErrClr	Acknowledgment of the present error	733	S32	1	0	ENUM	-
10	VADrtPriMod*	Prioritizing of control of derating	730	S32	1	0	ENUM	-
12	WGraMod*	Activates the active power gradient	725	S32	1	0	ENUM	-
14	WGra*	Active power gradient	726	S32	1000	0	FIX3	pu/s
16	VArGraMod*	Activates the reactive power gradient	727	S32	1	0	ENUM	-
18	VArGra*	Reactive power gradient	728	S32	1000	0	FIX3	pu/s
20	ErrClr.ProErr	Acknowledging of safety-relevant errors	7211	S32	1	0	ENUM	-
22	GriMng.InvVArMod*	Grid management services: specification of reactive power setpoint by the inverter	6088	S32	1	0	ENUM	-
26	Bfp.Ena	Activation of the function Backfeed Power	7372	S32	1	0	ENUM	-
28	GriMng.VolNomMod	Grid management services: setpoint via AC voltage	7449	S32	1	0	ENUM	-
30	InstFunc	Activation of installation functions	7469	S32	1	0	ENUM	-
34	PvGnd.OpnRemGfdi	Open the Remote GFDI	7682	S32	1	0	ENUM	-

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
36	StbySfCapacMod	Activation of the standby mode of the sine filter capacitor	7913	S32	1	0	ENUM	-
38	Aid.Mod*	Activation of active islanding detection	709	S32	1	0	ENUM	-
40	VCtl.Hi1Lim*	Monitoring the grid voltage: upper switch-off limit 1	446	S16	100	0	FIX2	pu
41	VCtl.Hi1LimTm*	Monitoring the grid voltage: waiting time upper switch-off limit 1	451	U32	1	0	FIX0	ms
43	VCtl.Hi2Lim*	Monitoring the grid voltage: upper switch-off limit 2	447	S16	100	0	FIX2	pu
44	VCtl.Hi2LimTm*	Monitoring the grid voltage: waiting time upper switch-off limit 2	452	U32	1	0	FIX0	ms
46	VCtl.Lo1Lim*	Monitoring the grid voltage: lower switch-off limit 1	456	S16	100	0	FIX2	pu
47	VCtl.Lo1LimTm*	Monitoring the grid voltage: waiting time lower switch-off limit 1	461	U32	1	0	FIX0	ms
49	VCtl.Lo2Lim*	Monitoring the grid voltage: lower switch-off limit 2	457	S16	100	0	FIX2	pu
50	VCtl.Lo2LimTm*	Monitoring the grid voltage: waiting time lower switch-off limit 2	462	U32	1	0	FIX0	ms
52	VCtl.Lo3Lim*	Monitoring the grid voltage: lower switch-off limit 3	458	S16	100	0	FIX2	pu
53	VCtl.Lo3LimTm*	Monitoring the grid voltage: waiting time lower switch-off limit 3	463	U32	1	0	FIX0	ms
55	VCtl.OpMaxNom*	Monitoring the grid voltage: upper switch-on limit	444	S16	100	0	FIX2	pu
56	VCtl.OpMinNom*	Monitoring the grid voltage: lower switch-on limit	445	S16	100	0	FIX2	pu
57	HzCtl.Hi1Lim*	Monitoring the grid frequency: upper switch-off limit 1	468	U16	100	0	FIX2	Hz
58	HzCtl.Hi1LimTm*	Monitoring the grid frequency: waiting time upper switch-off limit 1	474	U32	1	0	FIX0	ms
60	HzCtl.Hi2Lim*	Monitoring the grid frequency: upper switch-off limit 2	469	U16	100	0	FIX2	Hz

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
61	HzCtl.Hi2LimTm*	Monitoring the grid frequency: waiting time upper switch-off limit 2	475	U32	1	0	FIX0	ms
63	HzCtl.Lo1Lim*	Monitoring the grid frequency: lower switch-off limit 1	480	U16	100	0	FIX2	Hz
64	HzCtl.Lo1LimTm*	Monitoring the grid frequency: waiting time lower switch-off limit 1	486	U32	1	0	FIX0	ms
66	HzCtl.Lo2Lim*	Monitoring the grid frequency: lower switch-off limit 2	481	U16	100	0	FIX2	Hz
67	HzCtl.Lo2LimTm*	Monitoring the grid frequency: waiting time lower switch-off limit 2	487	U32	1	0	FIX0	ms
69	HzCtl.OpMaxNo m*	Monitoring the grid frequency: upper switch-on limit	466	U16	100	0	FIX2	Hz
70	HzCtl.OpMinNo m*	Monitoring the grid frequency: lower switch-on limit	467	U16	1	0	FIX0	Hz
71	WGraStr*	Start-up ramp for active power	7523	U16	1000	0	FIX3	pu/s
73	VArCtlVol.LoVolR ef3*	AC voltage-dependent reactive power control: reference point 3 (below reference point 1)	7523	S16	1000	0	FIX3	pu
74	VArCtlVol.LoVolR ef2*	AC voltage-dependent reactive power control: reference point 2 (below reference point 1)	6050	S16	1000	0	FIX3	pu
75	VArCtlVol.LoVolR ef1HiVolRef1*	AC voltage-dependent reactive power control: reference point 1	6048	S16	1000	0	FIX3	pu
76	VArCtlVol.HiVolR ef2*	AC voltage-dependent reactive power control: reference point 2 (via reference point 1)	6052	S16	1000	0	FIX3	pu
77	VArCtlVol.HiVolR ef3*	AC voltage-dependent reactive power control: reference point 3 (via reference point 1)	6053	S16	1000	0	FIX3	pu
78	VArCtlVol.LoGra3 *	AC voltage-dependent reactive power control: gradient 3 (from reference point 3)	6058	S16	100	0	FIX2	pu
79	VArCtlVol.LoGra2 *	AC voltage-dependent reactive power control: gradient 2 (from reference point 2)	6056	S16	100	0	FIX2	pu

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
80	VArCtlVol.LoGra1*	AC voltage-dependent reactive power control: gradient 1 (from reference point 1)	6054	S16	100	0	FIX2	pu
81	VArCtlVol.HiGra1*	AC voltage-dependent reactive power control: gradient 1 (from reference point 1)	6055	S16	100	0	FIX2	pu
82	VArCtlVol.HiGra2*	AC voltage-dependent reactive power control: gradient 2 (from reference point 2)	6057	S16	100	0	FIX2	pu
83	VArCtlVol.HiGra3*	AC voltage-dependent reactive power control: gradient 3 (from reference point 3)	6059	S16	100	0	FIX2	pu
84	WCtlHzMod	Activation of frequency-dependent active power control: To enable settings for this method via Modbus, the parameter WCtlHz.CfgMod must be set to W via the user interface.	361	S32	1	0	ENUM	-
86	WCtlHz.Hz1	Frequency-dependent active power control: cutoff frequency of the 1st frequency band	364	U16	100	0	FIX2	Hz
87	WCtlHz.Hz2	Frequency-dependent active power control: cutoff frequency of the 2st frequency band	365	U16	100	0	FIX2	Hz
88	WCtlHz.Hz3	Frequency-dependent active power control: cutoff frequency of the 3st frequency band	366	U16	100	0	FIX2	Hz
89	WCtlHz.Hz4	Frequency-dependent active power control: cutoff frequency of the 4st frequency band	708	U16	100	0	FIX2	Hz
90	WCtlHz.W2*	Frequency-dependent active power control: power referring to the inverter at the beginning of the second area	6584	S16	1000	0	FIX3	pu
91	WCtlHz.W3*	Frequency-dependent active power control: power referring to the inverter at the beginning of the third area	6586	S16	1000	0	FIX3	pu

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
92	WCtHz.W4*	Frequency-dependent active power control: power referring to the inverter at the beginning of the fourth area	6588	S16	1000	0	FIX3	pu
93	WCtIVol.Ena*	Activation of AC voltage-dependent active power control	8156	S32	1	0	ENUM	-
95	WCtIVol.Vol1*	AC voltage-dependent active power control: voltage at reference point 1	8160	S16	10000	0	FIX4	pu
96	WCtIVol.Vol2*	AC voltage-dependent active power control: voltage at reference point 2	8162	S16	10000	0	FIX4	pu
97	WCtIVol.Vol3*	AC voltage-dependent active power control: voltage at reference point 3	8164	S16	10000	0	FIX4	pu
98	WCtIVol.Vol4*	AC voltage-dependent active power control: voltage at reference point 4	8166	S16	10000	0	FIX4	pu
99	WCtIVol.W1*	AC voltage-dependent active power control: active power at reference point 1	8168	S16	100	0	FIX2	%
100	WCtIVol.W2*	AC voltage-dependent active power control: active power at reference point 2	8170	S16	100	0	FIX2	%
101	WCtIVol.W3*	AC voltage-dependent active power control: active power at reference point 3	8172	S16	100	0	FIX2	%
102	WCtIVol.W4*	AC voltage-dependent active power control: active power at reference point 4	8174	S16	100	0	FIX2	%
103 - 120	Reserved	-	-	-	-	-	-	-
121	WGraFlb	Active power gradient for the control in the event of a loss of communication	8568	S32	1000	0	FIX3	pu/s
123	VArGraFlb	Reactive power gradient for the control during a loss of communication	8572	S32	1000	0	FIX3	pu/s
125	VArCtIVol.VolPsNomHiLim	AC voltage-dependent reactive power control: upper limit for feedback voltage	8497	S16	10000	0	FIX4	pu

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
126	VArCtlVol.VolPsNomLoLim	AC voltage-dependent reactive power control: lower limit for feedback voltage	8499	S16	10000	0	FIX4	pu
127	ExlConn	External release for connection to the grid	8947	S32	1	0	ENUM	-
129	GriMng.ExtDFlbErrClr	Grid management services: Extended fallback operation, acknowledging the error	8994	S32	1	0	ENUM	-
131	PvGnd.RisIsoWarnLim	Limiting value for the warning of the insulation resistance	6640	U16	10	0	FIX1	kohm
132	PvGnd.RisIsoErrLim	Limiting value for the error of the insulation resistance	6642	U16	10	0	FIX1	kohm
133-134	Reserved	-	-	-	-	-	-	-
143	PvGnd.OpnRemGfdIsoMon	Activation of the insulation measurement with manually opened GFDI	9283	S32	1	0	ENUM	-

Related ENUM Codes

The following table contains number codes which, as function coding in data format ENUM, are frequently used in the SMA Modbus profile. The column "Related text" contains the corresponding abbreviations.

Address	Channel	Unique ID	Access	Return code	Related text
0	InvOpMod	329	RW	973	—
				381	Stop
				309	Operation
2	RemRdy	331	RW	973	—
				308	Enabled
				303	Disabled
4	GriMng.VArMod	6080	RW	1387	VArCtlAnIn
				1388	PFCtlAnIn
				1072	VArCtlCom
				1075	PFCtlCom
				2270	AutoCom
				1071	VArCtlMan

Address	Channel	Unique ID	Access	Return code	Related text
6	GriMng.WMod	6078	RW	1074	PFCtlMan
				303	Off
				1390	WCtlAnIn
				1079	WCtlCom
				1077	WCtlMan
8	ErrClr	733	RW	303	Off
				973	—
10	VADrtPriMod	730	RW	26	Ackn
				16	VAr
12	WGrMod	725	RW	18	W
				973	—
16	VArGrMod	727	RW	308	Enable
				303	Disable
20	ErrClr.ProErr	7211	RW	973	—
				21118	Gfdi
				28	Aid
				2490	Pld
				21119	IsoBender
				21120	Frq
				21204	VCtl.LoLim
				21205	VCtl.HiLim
				20907	VCtl.PkLim
				21121	All
22	GriMng.InvVArMod	6088	RW	303	Off
				21206	VArCtlVol
				21207	VArCtlVolPi
				21208	PFCtlW
26	Bfp.Ena	7372	RW	973	—
				308	Enable
				303	Disable
28	GriMng.VolNomMod	7449	RW	21485	VolNomCtlCom
				21486	VolNomCtlMan
				303	Off
30	InstFunc	7469	RW	973	—
				21488	Systemreset Acc, Dcc, Rio, Cont
34	PvGnd.OpnRemGfdi	7682	RW	973	—
				308	Enable
				303	Disable
36	StbySfCapacMod	7913	RW	973	—
				21679	Standby with capacitor

Address	Channel	Unique ID	Access	Return code	Related text
				21680	Standby without capacitor
38	Aid.Mod	709	RW	973	—
84	WCtlHzMod	361	RW	308	Enable
93	WCtlVol.Ena	8156	RW	303	Disable
127	ExlConn	8947	RW		
129	GriMng.ExtFlbErrClr	8994	RW	973	—
				26	Ackn
143	PvGnd.OpnRemGfdils oMon	9283	RW	973	—
				308	Enable
				303	Disable

5.3.2 Input Register

In the following table, you can find the measured values supported by the inverter, which you can access under unit ID = 3 and function code 0x04 (see Section 3.6.1 "Unit ID's", page 11).

Measured Values

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
0	DcMs.Vol	DC voltage DC plus to DC minus	597	S32	10	0	FIX1	V
2	DcMs.Amp.Stk1	DC current inverter bridge 1	600	S32	1	0	FIX0	A
4	DcMs.Amp.Stk2	DC current inverter bridge 2	601	S32	1	0	FIX0	A
6	DcMs.Amp.Stk3	DC current inverter bridge 3	602	S32	1	0	FIX0	A
8	DcMs.TotAmp	DC current, total	6146	S32	1	0	FIX0	A
10	DcMs.TotWatt	DC power, total	603	S32	1	0	FIX0	kW
12	InvMs.DclVol.Stk1	DC voltage of the inverter bridge 1	616	S32	10	0	FIX1	V
14	InvMs.DclVol.Stk2	DC voltage of the inverter bridge 2	617	S32	10	0	FIX1	V
16	InvMs.DclVol.Stk3	DC voltage of the inverter bridge 3	618	S32	10	0	FIX1	V
18	InvMs.TotA.Phs A	AC current L1, total	408	S32	1	0	FIX0	A
20	InvMs.TotA.Phs B	AC current L2, total	409	S32	1	0	FIX0	A
22	InvMs.TotA.Phs C	AC current L3, total	410	S32	1	0	FIX0	A
24	InvMs.PF	Power factor (cos phi)	404	S32	1000 0	0	FIX4	-
26	InvMs.TotVA	Apparent power, total	401	S32	1	0	FIX0	kVA
28	InvMs.TotW	Active power, total	402	S32	1	0	FIX0	kW
30	InvMs.TotVA _r	Reactive power, total	403	S32	1	0	FIX0	kVA _r
32	GriMs.V.PhsAB	Line-to-line voltage L1-L2, utility grid	405	S32	10	0	FIX1	V
34	GriMs.V.PhsBC	Line-to-line voltage L2-L3, utility grid	406	S32	10	0	FIX1	V
36	GriMs.V.PhsCA	Line-to-line voltage L3-L1, utility grid	407	S32	10	0	FIX1	V

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
38	GriMs.Hz	Grid frequency	607	S32	100	0	FIX2	Hz
40	TmpCab.Acc	Temperature in the AC range	751	S32	10	0	FIX1	°C
42	TmpCab.Dcc	Temperature in the DC range	750	S32	10	0	FIX1	°C
44	TmpCab.Rio	Temperature in the electronics area	752	S32	10	0	FIX1	°C
46	TmpCab.Max	Temperature in interior, maximum value	6130	S32	10	0	FIX1	°C
48	TmpStk.IgbtMax	Temperature of the inverter bridge (switch), maximum value	6100	S32	10	0	FIX1	°C
50	TmpStk.PcbMax	Temperature of the inverter bridge (assembly), maximum value	6099	S32	10	0	FIX1	°C
52	TmpExl	External temperature	759	S32	10	0	FIX1	°C
54	TmpTrf	Temperature of the MV transformer <i>Not available for MVPS2500-S-AU / MVPS 2750-S-AU / MVPS 3000-S-AU / MVPS5000-S-AU / MVPS 5500-S-AU / MVPS 6000-S-AU</i>	6107	S32	10	0	FIX1	°C
56	DcSw1Stt	Switching status of the DC load-break switch 1	611	S32	1	0	ENUM	-
58	DcSw2Stt	Switching status of the DC load-break switch 2	721	S32	1	0	ENUM	-
60	DcSw3Stt	Switching status of the DC load-break switch 3	722	S32	1	0	ENUM	-
62	AcSwStt	Switching status of the AC circuit breaker	6202	S32	1	0	ENUM	-
66	CapacSwStt	Switching state of the capacitor contactor	6708	S32	1	0	ENUM	-
68	PrchrgSwStt	Switching status of the pre-charging contactor	6707	S32	1	0	ENUM	-
70	Cnt.TotFeedTm	Feed-in time	6779	U32	1	0	FIX0	s
72	Cnt.TotOpTm	Operating time	6777	U32	1	0	FIX0	s
74	Cnt.TotWatthIn	DC energy drawn, total	7119	S32	100	0	FIX2	MWh

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
76	Cnt.TotWhOut	AC energy fed in, total	7118	S32	100	0	FIX2	MWh
78	Cnt.WatthIn	DC energy drawn that day	6771	S32	100	0	FIX2	MWh
80	Cnt.WhOut	AC energy fed in that day	6767	S32	100	0	FIX2	MWh
82	Cnt.FanCab1Tm	Operating time of the interior fan 1	6793	U32	1	0	FIX0	s
84	Cnt.FanCab2Tm	Operating time of the interior fan 2	6795	U32	1	0	FIX0	s
86	Cnt.FanStkTm	Operating time of the inverter bridge fan	6791	U32	1	0	FIX0	s
88	Cnt.HtCabTm	Operating time of the interior heater	6797	U32	1	0	FIX0	s
90	Cnt.HtLoExlTm Tm	Operating time of the -40°C interior heater	6799	U32	1	0	FIX0	s
92	ErrLcn	Error location of the present error	6968	U32	1	0	ENUM	-
94	ErrStt	Error status of the inverter	328	S32	1	0	ENUM	-
96	ErrNo	Error number of the present error	823	U32	1	0	FIX0	-
98	OpStt	Operating state of the inverter	332	S32	1	0	ENUM	-
100	Rio.KeySw	Switch status of the start-stop key switch	327	S32	1	0	ENUM	-
102	WaitGriTm	Waiting time for grid connection	614	S32	10	0	FIX1	s
104	DevInf.SerNo	Serial number of the inverter	6763	U32	1	0	FIX0	-
106	GfdiSwStt	Switching status of the GFDI	6706	S32	1	0	ENUM	-
108	WSpt	Active power, setpoint	320	S32	1	0	FIX0	kW
110	VAMaxSpt	Maximum apparent power, setpoint	665	S32	10	0	FIX1	kVA
112	VArSpt	Reactive power, setpoint	321	S32	1	0	FIX0	kVAr
114	PFSpt	Power factor cos (phi), setpoint	322	S32	1000 0	0	FIX4	-
116	TrfPro.Pres	Fast stop hermetic protection, pressure	7029	S32	1	0	ENUM	-
118	TrfPro.TmpTrp	Fast stop hermetic protection, temperature	506	S32	1	0	ENUM	-

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
120	TrfPro.GasOilLevel	Hermetic protection, gas fill level	7170	S32	1	0	ENUM	-
122	TrfPro.TmpWrn	Hermetic pressure, temperature warning	7168	S32	1	0	ENUM	-
124	TmpStk1.lgbt	Temperature of inverter bridge 1 (switch)	756	S32	10	0	FIX1	°C
126	TmpStk2.lgbt	Temperature of inverter bridge 2 (switch)	757	S32	10	0	FIX1	°C
128	TmpStk3.lgbt	Temperature of inverter bridge 3 (switch)	758	S32	10	0	FIX1	°C
130	TmpStk1.Pcb	Maximum temperature of inverter bridge 1 (assembly)	753	S32	10	0	FIX1	°C
132	TmpStk2.Pcb	Maximum temperature of inverter bridge 2 (assembly)	754	S32	10	0	FIX1	°C
134	TmpStk3.Pcb	Maximum temperature of inverter bridge 3 (assembly)	755	S32	10	0	FIX1	°C
136	Cnt.YstdWhOut	AC energy fed in the previous day	7073	S32	100	0	FIX2	MWh
138	Cpu1UpTime	CPU running time since system start	7225	U32	1	0	FIX0	s
140	PvGnd.Rislso	Insulation resistance	6644	S32	10	0	FIX1	kohm
142	PresTrf	MV transformer pressure <i>Not available for MVPS2500-S-AU / MVPS 2750-S-AU / MVPS 3000-S-AU / MVPS5000-S-AU / MVPS 5500-S-AU / MVPS 6000-S-AU</i>	7249	S32	100	0	FIX2	-
144	PresTrf.ErrStt	Error status of MV transformer pressure monitoring	7253	S32	0	0	ENUM	-
146	BfpBits	Status of the function Backfeed Power	7383	U16	1	0	FIX0	-
147	DcMs.BfpAmp	Backfeed Power: DC current used by customer loads	7382	S32	10	0	FIX1	A
149	GriMs.Vol.PsNm	Normalized AC voltage, positive sequence	7233	S32	1000 0	0	FIX4	-

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
151	Cnt.TotDcWhOut	DC energy fed in, total	7300	S32	100	0	FIX2	MWh
153	Cnt.TotAcWhIn	AC energy drawn, total	7182	S32	100	0	FIX2	MWh
155	Cnt.DcWhOut	DC energy fed in that day	7302	S32	100	0	FIX2	MWh
157	Cnt.AcWhIn	AC energy drawn that day	7180	S32	100	0	FIX2	MWh
159	DclVolSpt	DC voltage, setpoint	7488	S32	10	0	FIX1	V
161	Reserved	-	-	-	-	-	-	-
163	VolNomSpt	AC voltage-dependent reactive power control: setpoint AC voltage	7000	S32	1000 0	0	FIX4	pu
167	AuxCtl.LifeSign	Life sign with valid setpoints	7355	U16	1	0	FIX0	-
168	InvPwrVolTyp	Inverter type	6338	S32	1	0	ENUM	-
170	HzNomSpt	Nominal frequency for frequency-dependent control of active power	7277	S32	100	0	FIX2	Hz
172	WAval	Currently available active power	7632	S32	1000 0	0	FIX4	pu
174	VArAval	Currently available reactive power	7633	S32	1000 0	0	FIX4	pu
176	DrtStt	Current derating status	830	S32	1	0	ENUM	-
178	PwrOffReas	Current cause of inverter disconnection	7221	S32	1	0	ENUM	-
180	DiagRmgTm	Time until the next inverter self test	7716	U32	1	0	FIX0	s
182	Rio.Din.FloatCtl.Warn	The float controller indicates a warning.	6382	U16	1	0	FIX0	-
183	Rio.Din.FloatCtl.Err	The float controller indicates an error	6383	U16	1	0	FIX0	-
184	WRtg	Nominal value of active power	318	S32	1	0	FIX0	kW
186	Gfdi.AmpPrc	Ground current measured at the GFDI	7986	S16	100	0	FIX2	A
187	Gfdi.AmpErr	Last measured ground current in case of an error	7988	S16	100	0	FIX2	A
188	ActErrNo1	Error number of the current error	8045	U32	1	0	FIX0	-

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
190	ActErrLcn1	Location where the current error was detected	8047	U32	1	0	ENUM	-
192	ActErrNo2	Error number of the current error	8048	U32	1	0	FIX0	-
194	ActErrLcn2	Location where the current error was detected	8050	U32	1	0	ENUM	-
196	ActErrNo3	Error number of the current error	8051	U32	1	0	FIX0	-
198	ActErrLcn3	Location where the current error was detected	8053	U32	1	0	ENUM	-
200	ActErrNo4	Error number of the current error	8054	U32	1	0	FIX0	-
202	ActErrLcn4	Location where the current error was detected	8056	U32	1	0	ENUM	-
204	ActErrNo5	Error number of the current error	8057	U32	1	0	FIX0	-
206	ActErrLcn5	Location where the current error was detected	8059	U32	1	0	ENUM	-
208	ActErrNo6	Error number of the current error	8060	U32	1	0	FIX0	-
210	ActErrLcn6	Location where the current error was detected	8062	U32	1	0	ENUM	-
212	ActErrNo7	Error number of the current error	8063	U32	1	0	FIX0	-
214	ActErrLcn7	Location where the current error was detected	8065	U32	1	0	ENUM	-
216	ActErrNo8	Error number of the current error	8066	U32	1	0	FIX0	-
218	ActErrLcn8	Location where the current error was detected	8068	U32	1	0	ENUM	-
220	ActErrNo9	Error number of the current error	8069	U32	1	0	FIX0	-
222	ActErrLcn9	Location where the current error was detected	8071	U32	1	0	ENUM	-
224	ActErrNo10	Error number of the current error	8072	U32	1	0	FIX0	-

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
226	ActErrLcn10	Location where the current error was detected	8074	U32	1	0	ENUM	-
227 - 267	Reserved	-	-	-	-	-	-	-
268	VARExlSpt.RefVal	Reference value for the external reactive power setpoint	8271	S16	1	0	FIX0	kvar
269	WExlSpt.RefVal	Reference value for the external active power setpoint	8272	S16	1	0	FIX0	kW
270	VARtg	Nominal value of apparent power	323	S32	1	0	FIX0	kVA
272	InvTyp	Inverter type	7359	S32	1	0	ENUM	-
274	VARtg	Nominal value of reactive power	319	S32	1	0	FIX0	kvar
276	DcMs.Vol.PosGnd	DC voltage of DC+ to ground	598	S32	10	0	FIX1	V
278	DcMs.Vol.NegGnd	DC voltage of DC- to ground	599	S32	10	0	FIX1	V
286	GriMng.ExtFlbStt	Grid management services: Extended fallback operation, status	8993	U32	1	0	ENUM	-
288	ErrRmgTm	Remaining error waiting period	7114	U32	1	0	FIX0	s
290	Cnt.DcSw1	Switch cycles of the DC load-break switch 1	6803	U32	1	0	FIX0	-
292	Cnt.DcSw2	Switch cycles of the DC load-break switch 2	6805	U32	1	0	FIX0	-
294	Cnt.DcSw3	Switch cycles of the DC load-break switch 3	6807	U32	1	0	FIX0	-
296	Cnt.AcSw	Switch cycles of the AC circuit breaker	6801	U32	1	0	FIX0	-
298-329	Reserved	-	-	-	-	-	-	-
330	Cnt.DcSw1Uvr	Switch cycles of the DC load-break switch 1, undervoltage release	9314	U32	1	0	FIX0	-

Address	Channel name	Explanation	Unique ID	Data type	Scaling	Offset	Data format	Unit
332	Cnt.DcSw2Uvr	Switch cycles of the DC load-break switch 2, undervoltage release	9316	U32	1	0	FIX0	-
334	Cnt.DcSw3Uvr	Switch cycles of the DC load-break switch 3, undervoltage release	9318	U32	1	0	FIX0	-
336	Cnt.AcSwUvr	Switch cycles of the AC circuit breaker, undervoltage release	9312	U32	1	0	FIX0	-
338	WSptMin	Active power, minimum setpoint	8253	S32	1	0	FIX0	kW
339	WSptMax	Active power, maximum setpoint	8537	S32	1	0	FIX0	kW
340	RIHmdtRio	Relative humidity in the electronics area	760	S16	10	0	FIX1	%
341	RIHmdtAcc	Relative humidity in the AC area	761	S16	10	0	FIX1	%
342	RIHmdtDcc	Relative humidity in the DC area	762	S16	10	0	FIX1	%
343	DrtExlStt	Derating due to outside temperature	690	S32	1	0	ENUM	-
345	HmdtCtl.TmCnt	Time counter to activate/deactivate dehumidification	9515	U32	1	0	FIX0	s
347	HmdtCtl.Stt	Status of dehumidification activation	9517	S32	1	0	ENUM	-

Related ENUM Codes

The following table contains number codes which, as function coding in data format ENUM, are frequently used in the SMA Modbus profile. The column "Related text" contains the corresponding abbreviations.

Address	Channel	Unique ID	Access	Return code	Related text
56	DcSw1Stt	611	RO	973	—
58	DcSw2Stt	721	RO	311	Open
60	DcSw3Stt	722	RO	51	Closed
62	AcSw1Stt	6202	RO		
66	CapacSwStt	6708	RO		
68	PrchrgSwStt	6707	RO		

Address	Channel	Unique ID	Access	Return code	Related text
92	ErrLcn	332	RO	0	
190	ActErrLcn1	8047	RO	238	AC contactor
194	ActErrLcn2	8050	RO	267	Inverter
198	ActErrLcn3	8053	RO	268	Insulation monitoring
202	ActErrLcn4	8056	RO	302	—
206	ActErrLcn5	8059	RO	422	Transformer
210	ActErrLcn6	8062	RO	465	DC disconnection unit
214	ActErrLcn7	8065	RO	512	Surge arrester
218	ActErrLcn8	8068	RO	1396	AC grid
222	ActErrLcn9	8071	RO	3278	SC30DST
226	ActErrLcn10	8074	RO	3279	SC30ACC
				3280	SC30DCC
				3281	SC30RIO
				3282	SC30CONT
				3283	SC30CONT CPU1
				3284	SC30CONT CPU2
				4260	DC pre-charge
				21006	SC30DST 1
				21007	SC30DST 2
				21008	SC30DST 3
				21009	SC30DST 1 EEPROM
				21010	SC30DST 2 EEPROM
				21011	SC30DST 3 EEPROM
				21012	SC30APP
				21013	SC30DCC EEPROM
				21014	SC30ACC EEPROM
				21015	SC30DCSHUNT
				21016	Bender
				21020	Pre-charge contactor
				21021	Sinusoidal filter capacitor
				21055	contactor
				21530	GFDI
					Pre-charging contactor sine-wave
				21534	filter capacitor
				21535	On-board power supply
				21638	Circuit breaker
				21677	SC30CONT FPGA
				21749	Float Controller
				21750	DC/DC Converter 1
				21751	DC/DC Converter 2
				21752	DC/DC Converter 3
				21753	DC/DC Converter 4
				21754	DC/DC Converter 5
				21757	DC/DC Converter 6
				21758	Battery management system
				21759	Battery management system 2
				21760	Battery management system 3
				21761	Battery management system 4
				21762	Battery management system 5
				21819	Battery management system 6
				21820	DC load-break switch 1
				21821	DC load-break switch 2
					DC load-break switch 3
94	ErrStt	328	RO	973	—
				307	OK
				1392	Error
98	OpStt	332	RO	1787	Init

Address	Channel	Unique ID	Access	Return code	Related text
				381	Stop
				1392	Error
				3522	Update
				3523	Reset
				1394	WaitAC
				3524	ConnectAC
				1393	WaitDC
				3525	ConnectDC
				3526	GridFeed
				3527	FRT
				3528	Standby
				3529	QonDemand
				3530	RampDown
				1469	ShutDown
				375	Selftest
				3533	IOTest
				22509	HumidityMng
100	Rio.KeySw	327	RO	973	—
				308	On
				303	Off
106	GfdiSwStt	6706	RO	973	—
				311	Open
				51	Closed
116	TrfPro.Pres	7029	RO	973	—
118	TrfPro.TmpTrp	506		307	OK
120	TrfPro.GasOilLev	7170		1392	Error
122	TrfPro.TmpWrn	7168			
144	PresTrf.ErrStt	7253			
168	InvPwrVolTyp	6338	RO	9299	Sunny Central 2200-US
				9417	Sunny Central 2500-EV
				9379	Sunny Central 1850-US
				9298	Sunny Central 2200
				9380	Sunny Central 2500-EV-US
				9387	Sunny Central 2750-EV
				9388	Sunny Central 2750-EV-US
				9412	Sunny Central 2475
				9413	Sunny Central 3000-EV
				9415	Sunny Central 1760-US
				9416	Sunny Central 2000-US
				9433	Sunny Central 2000-EV-US
				9457	Sunny Central 4000-UP
				9458	Sunny Central 4000-UP-US
				9461	Sunny Central 4200-U
				9462	Sunny Central 4200-UP-US
				9465	Sunny Central 4400-UP

Address	Channel	Unique ID	Access	Return code	Related text
				9466	Sunny Central 4400-UP-US
				9469	Sunny Central 4600-UP
				9470	Sunny Central 4600-UP-US
				19054	Sunny Central 2660-UP
				19055	Sunny Central 2660-UP-US
				19056	Sunny Central 2800-UP
				19057	Sunny Central 2800-UP-US
				19058	Sunny Central 2930-UP
				19059	Sunny Central 2930-UP-US
				19060	Sunny Central 3060-UP
				19061	Sunny Central 3060-UP-US
				22017	Sunny Central VARY UP (3S)
				22019	Sunny Central VARY UP-US (3S)
				22273	Sunny Central VARY UP (2S)
				22275	Sunny Central VARY UP-US (2S)
176	DrtStt	830	RO	973	—
				21585	Stk.DcAmplim
				21586	Frt
				21587	AmpGra
				21588	AMax
				21589	VAMax
				21590	WCtlLoHz
				21591	WCtlHz
				21592	WGraRecon
				21593	WGra
				21594	WMax
				21595	WMaxExt
				21596	VecLen
				21599	WGraStr
				21601	WCtlVol
				21602	Mvps.WRvLim
				21651	VArPrio
178	PwrOffReas	72214	RO	973	—
				21603	No Power Off Reason
				21604	Error: Critical Error, ProErr active
				21605	Error
				21606	Stop: Key Switch
				21607	Stop: Parameter InvOpmod
				21608	Stop: Stop External X440:3
				21609	Stop: Scada or PPC, Modbus
				21610	Stop: unspecified

Address	Channel	Unique ID	Access	Return code	Related text
				21612	Standby: Scada or PPC, Modbus
				21613	Standby: AC Synchronization
				21614	Standby: Low DC Power
				21615	Standby: External Grid Error
				21616	Standby: Power Monitoring Module
				21617	Standby: Parameter RemRdy
				21618	Standby External X440:7
				21619	Standby: unspecified
				21620	WaitAc
				21621	WaitDc: DC Voltage
				21622	WaitDc: Bender
				21623	WaitDc: DC precharge waiting period
				21624	Selftest active
				21625	IO Test active
				21626	Low Power Set Point
272	InvTyp			21983	PV inverter
				21984	Battery inverter
				21985	PV and battery inverter
286	GriMng.ExtdFlbStt			307	OK
				22232	Timeout, inactive
				22233	Timeout, 1 hour
				22234	Timeout, 30 seconds
				22235	Fallback and error
343	DrtExlStt			973	—
				22463	No VAMax derating
				22464	Derating VAMax MVPS communication
				22465	Derating VAMax external temperature
				22466	Derating VAMax transformer temperature
347	HmdtCtl.Stt			973	—
				308	On
				303	Off

Explanation for the current derating statuses

TExt	Explanation
—	Derating not active
Stk.DcAmpLim	The DC current limitation has been reached
Frt	The inverter is in the "FRT" operating state and supports the utility grid via dynamic grid support
AmpGra	The active power limitation is active because a gradual increase in the feed-in power is being performed.
AMax	The maximum AC current has been reached.
VAMax	The maximum AC voltage has been reached.

TExt	Explanation
WCtlLoHz	Active power limitation is active due to underfrequency.
WCtlHz	Active power limitation is active due to overfrequency.
WGraRecon	Active power limitation is active because of a grid error during the gradual increase in the feed-in power.
WGra	Active power limitation is active because of a gradual increase in the feed-in power in MPP operation.
WMax	The maximum AC power has been reached.
WMaxExt	The maximum AC power has been reached. Here, the limitation is specified via an external set value.
VeLen	The active power has reduced due to the DC voltage.
WGraStr	Active power limitation is active because of a gradual increase in the feed-in power in MPP operation.
WCtlVol	Active power limitation is active due to the grid voltage.
Mvps.WRvLim	Active power limitation is active to protect the MV transformer in the MV Power Station.
VArPrio	The active power is reduced due to required reactive power feed-in.

Explanation for the cause of the inverter disconnecting and corrective measures

TExt	Explanation	CORRECTIVE MEASURES
Invalid	The inverter has not initialized.	–
No Power Off Reason	A reason for the disconnection could not be determined.	Wait briefly for the inverter reaction.
Error: Critical Error, ProErr active	A safety critical error is present. The error number is to be found in the instantaneous value ErrNo and ProErr .	- Check the cause of error. - Acknowledge the error once the cause has been remedied.
Error	An error is present. The error number is to be found in the instantaneous value ErrNo .	- Check the cause of error. - Acknowledge the error once the cause has been remedied.
Stop: Key Switch	The key switch is in the Off position.	Switch the key switch on.
Stop: Parameter InvOpMod	The parameter InvOpMod is not set to Operation .	Set the parameter InvOpMod to Operation.
Stop: Stop External X440:3	The signal to stop the inverter was sent via digital input X440:3 .	Measure input X440:3 and check the cable if necessary.
Stop: Scada or PPC, Modbus	The last request via Modbus was an inverter stop request.	Request that the inverter be started via Modbus.
Stop: unspecified	A reason for the inverter stopping could not be determined.	- Wait 30 seconds. - Restart the inverter. If the error is still present after startup, contact Service.
Standby: Scada or PPC, Modbus	The last request via Modbus was an inverter standby request.	Request that the inverter be started via Modbus.
Standby: AC Synchronization	Synchronization with the AC grid is not possible.	Check the voltages at the inverter input.
Standby: Low DC Power	The inverter is in standby because the DC voltage is too low.	Wait for the DC voltage to increase.
Standby: External Grid Error	An external grid error was sent via Modbus.	- Delete the external grid error. - Acknowledge the error via Modbus.
Standby: Power Monitoring Module	The buffer module or the inverter-internal diagnostics requested standby mode.	- Check the voltage of the buffer module. - Check the external power specification.
Standby: Parameter RemRdy	The RemRdy parameter is set to Disable .	Set the parameter RemRdy to Disable.
Standby: Standby External X440:7	The signal to switch the inverter to standby was sent via digital input X440:7 .	Measure input X440:7 and check the cable if necessary.

TExt	Explanation	CORRECTIVE MEASURES
Standby: unspecified	A reason for the inverter switching to standby could not be determined.	• Wait 30 seconds. • Restart the inverter. If the error is still present after startup, contact Service.
WaitAc	The inverter is waiting for a valid AC grid.	• Wait until the waiting time for grid connection specified in the instantaneous value WaitGriTm has elapsed. • Check the AC voltage and grid limits in the parameters VCtl.xxx. If there a reason for the inverter switching off cannot be found, contact Service.
WaitDc: DC Voltage	The DC voltage is outside of the defined range.	• Wait or change DC voltage.
WaitDc: Bender	The insulation monitoring device is measuring the insulation resistance of the PV system.	• Wait until the insulation resistance measurement has been completed.
WaitDc: DC precharge waiting period	The waiting time for DC pre-charging has not elapsed.	• Wait until the waiting time for DC pre-charging has elapsed.
Selftest active	The inverter is performing a self test.	• Wait until the self test has been completed.
IO Test active	The inverter is in test mode.	• Please contact SMA Service.
Low Power SetPoint	Low power values are preventing the inverter from starting up.	• Increase the setpoints for active power in the parameter WSpt and reactive power in parameter VARspt.

5.4 Zone Monitoring Parameter (Unit ID = 32, 33 and 34)

In the following table, you can find the measured values and parameters supported by zone monitoring, which you can access under unit ID = 32, 33 and 34 (see Section 3.6.1 "Unit ID's", page 11).

The assignment of the DC inputs to the Modbus registers depends on the inverter configuration:

Sunny Central with DC fuse switches

Address	Unit ID 32: left DC module DC input	Unit ID 33: right DC module DC input
55	X21	X41
56	X22	X42
57	X23	X43
58	X24	X44
59	X25	X45
60	X26	X46
61	X27	X47
62	X28	X48
63	X29	X49
64	X30	X50
65	X31	X51
66	X32	X52
67	X33*	X53*
68	X34*	X54*
69	X35*	X55*
70	X36*	X56*

* These DC inputs are only available in the Sunny Central 1800-US, Sunny Central 2200-US and Sunny Central UP-US.

Usable connection tabs

Option	Unused input straps	Unused register
24 fused inputs with switches for the Sunny Central UP-US	Input straps X33 to X36 as well as X53 to X56 cannot be used with the option "24 fused inputs with switches".	Registers 67, 68, 69 and 70 for Unit ID 32 and for Unit ID 33 cannot be used with the option "24 fused inputs with switches".
28 fused inputs with switches for the Sunny Central UP-US	Input straps X53 to X56 cannot be used with the option "28 fused inputs with switches".	Registers 67, 68, 69 and 70 for Unit ID 33 cannot be used with the option "28 fused inputs with switches".

Sunny Central without DC fuse switches: Two-pole fused and insulated system with 24 DC inputs

Address	Unit ID 32: left DC module DC input	Unit ID 33: right DC module DC input
---------	--	---

Address	Unit ID 32: left DC module DC input	Unit ID 33: right DC module DC input
55	X61	X73
56	X62	X74
57	X63	X75
58	X64	X76
59	X65	X77
60	X66	X78
61	X67	X79
62	X68	X80
63	X69	X81
64	X70	X82
65	X71	X83
66	X72	X84
67 to 70	Not used	Not used

Sunny Central without DC fuse switches: One-pole fused, grounded and insulated system with 24 DC inputs

Address	Unit ID 32: left DC module DC input	Unit ID 33: right DC module DC input
55	X21	X41
56	X22	X42
57	X23	X43
58	X24	X44
59	X25	X45
60	X26	X46
61	X31	X51
62	X32	X52
63	X33	X53
64	X34	X54
65	X35	X55
66	X36	X56
67 to 70	Not used	Not used

Sunny Central without DC fuse switches: Single-pole fused, grounded and insulated system with 32 DC inputs

Address	Unit ID 32: left DC module DC input	Unit ID 33: right DC module DC input
55	X21	X41
56	X22	X42
57	X23	X43
58	X24	X44
59	X25	X45
60	X26	X46
61	X27	X47
62	X28	X48
63	X31	X51
64	X32	X52
65	X33	X53
66	X34	X54
67	X35	X55
68	X36	X56
69	X37	X57
70	X38	X58

Usable connection tabs

Option	Unused input straps	Unused register
28 fused inputs	Input straps X33, X36 as well as X53 and X56 cannot be used with the option "28 fused inputs".	Registers 65 and 68 for Unit ID 32 and for Unit ID 33 cannot be used with the option "28 fused inputs with switches".

6 Modbus Profiles for External Devices

With the Modbus device profile for external devices, Modbus devices (subordinate devices) of other manufacturers can be integrated in the internal LAN1 network of the inverter. With the Modbus device profile, Modbus devices of another manufacturer behave like Modbus devices from the SMA Solar Technology AG when its data is obtained via the communication interface of the inverter. If Modbus devices of other manufacturers are integrated into the station subdistribution of the MV station, these Modbus devices must also be integrated in the internal LAN1 network of the inverter in order to be accessible.

The Modbus device profile is an XML file in which, for each channel tag, one Modbus register of a slave device is assigned to any Modbus address. For the assignments, an address range from 0 to 65535 is available in each case. Currently, various Moxa IO modules from the ioLogik series are supported (see Section 9.1 "Unit ID's", page 57). The relevant XML templates can be acquired, adapted and imported from SMA Solar Technology AG.

The stored assignments are queried in a definable interval (default: 1000 ms).

Currently, the Modbus device profile contains register assignments for the following devices:

- "Moxa ioLogik E1210-T" (see Section 6.2 "Moxa ioLogik E1210-T - Assignment Table", page 51)
- "Moxa ioLogik E1240-T" (see Section 6.3 "Moxa ioLogik E1240-T - Assignment Table", page 51)
- "Moxa ioLogik E1242-T" (see Section 6.4 "Moxa ioLogik E1242-T - Assignment Table", page 52)
- "Moxa ioLogik E1260-T" (see Section 6.5 "Moxa ioLogik E1260-T - Assignment Table", page 53)

6.1 Structure of the XML File for the Modbus Device Profile

The Modbus device profile is created in the file *.mmp.

The basic structure of the XML file is as follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<ModbusMasterProfile>
  <ProfileVersion version="514">
  <ModbusDevices>
    <Device name="aaa" type="bbb" description="ccc">
      <Modbus ipaddress="192.168.100.16" port="502" dst_unitid="16"/>
      <Mapping src_unitid="1">
        <Channel />
        ...
      </Mapping>
    </Device>
  </ModbusDevices>
</ModbusMasterProfile>
```

Legend for XML tags and attributes:

The term "source register" in the following table stands for a specific Modbus register of the slave device, which is used as source for the assignment.

XML tag / attribute	Explanation
<ModbusMasterProfile>	The Modbus device profile is defined within this XML structure.
</ModbusMasterProfile>	

XML tag / attribute	Explanation
<ModbusDevices> </ModbusDevices>	The Modbus device profile is defined within this XML structure.
Device name="aaa"	Here you can assign any term for the device name. The XML template contains an entry appropriate for the device.
type="bbb"	Here you can assign any term for the device type. The XML template contains an entry appropriate for the device.
description="ccc"	Here you can assign any term for the use. The XML template contains an entry appropriate for the device.
Modbus ipaddress ="192.168.100.16"	IPv4 address of the slave device. The IP addresses can be assigned in the range from 192.168.100.16 to 192.168.100.31.
port="502"	Modbus communication port of the slave device. The default setting is 502.
dst_unitid	Unit ID under which the data read by the slave device can be obtained from the communication interface of the inverter.
src_unitid	Unit ID (8 bit) of the slave device (see manufacturer datasheet).
<Mapping> </ Mapping >	Within this XML structure, the profiles of the different channels are defined.
<Channel />	Within every channel tag, one source register of a slave device is assigned to the Modbus profile.

The following attributes can be assigned to each channel:

XML attribute	Explanation
name="ddd"	Here you can assign any name to the assignment.
tagid_index	Numerical index as suffix to the name of a channel (e.g., Aln1, Aln2, Aln3). If no index is required, then this attribute can remain empty.
source	Decimal address of the source register (see attribute "destination"). You can find possible "source" values in the datasheet of the manufacturer of each Modbus device.
destination	Decimal address (0 to 65535) of the Modbus register to which the value is assigned (destination register). Observe the length (in WORD) of the source register (see attribute "size").
datatype	Data format of the saved value, e.g., DT = date, FIXn = output with n decimal places, TEMP = output as temperature (see Section 0)
type	Type of the register, e.g., Holding (see Section 3.7)
size	Length of the source register (in WORD) (1 = 16 or 2 = 32), depending on register type. Holding or input register: 1, 2 Coil: 1 Depending on the length of the source register (1 or 2 WORD), the target addresses should be set for the status register and quality register following the "destination" register, e.g., destination = 1000 (size = 2), status = 1002, quality = 1003 + 1004, ensuring that they do not overlap with the subsequent registers.

XML attribute	Explanation
swap	Definition of sequence of the Modbus register for data transmission 0 - first the high byte and then the low byte of the Modbus register 1 - first the low byte and then the high byte of the Modbus register
scale	Specification of scaling factor with which the data must be converted
offset	Signed offset value which is to be added to the source data. The default setting is 0.
min	Lower value of the permitted range for valid values of the channel
max	Upper value of the permitted range for valid values of the channel
representation	Data format of the saved value, e.g., DT = date, FIXn = output with n decimal places, TEMP = output as temperature (see Section 0)
unit	Displayed unit of the channel

Modbus exceptions

For Modbus exceptions, see "Modbus Application Protocol Specification" at <http://www.modbus.org/specs.php>.

6.2 Moxa ioLogik E1210-T – Assignment Table

In this section you will find a table with an overview of the Modbus registers that are saved in the Modbus master profile for the "Moxa ioLogik E1210-T". In accordance with the XML structure, each line of the table shows the attribute values of a Modbus register assignment. The attribute values in the columns "source", "type" and "size" correspond to the standard values of the "Moxa ioLogik E1210-T".

Changed settings of the Moxa ioLogik E1210-T

If you change the settings (e.g., ip, description, etc.) of the "Moxa ioLogik E1210-T", you also have to change the attribute values of the Modbus master profile.

The following channels are defined in the profile file:

Channel/name	source	destination	type	datatype	size (WORD)	swap	scale	offset	min	max	representation	unit
"DI_status_0" to "DI_status_15"	0 to 15	0 to 15	Discrete	Bool	1	0	1	0	0	1	FIX0	-

6.3 Moxa ioLogik E1240-T – Assignment Table

In this section you will find a table with an overview of the Modbus registers that are saved in the Modbus master profile for the "Moxa ioLogik E1240-T". In accordance with the XML structure, each line of the table shows the attribute values of a Modbus register assignment. The attribute values in the columns "source", "type" and "size" correspond to the standard values of the "Moxa ioLogik E1240-T".

Changed settings of the Moxa ioLogik E1240-T

If you change the settings (e.g., ip, description, etc.) of the "Moxa ioLogik E1240-T", you also have to change the attribute values of the Modbus master profile.

The following channels are defined in the profile file:

Channel/name	source	destination	type	datatype	size (WORD)	swap	scale	offset	min	max	representation	unit
"Aln_Raw_0" to "Aln_Raw_7"	0 to 7	0 to 7	Input	U16	1	0	1	0	0	65535	FIX0	-
"Aln_Float_0" to "Aln_Float_7"	8 to 22	8 to 22	Input	F32	2	0	1	0	4.0	20.0	FIX3	mA

6.4 Moxa ioLogik E1242-T – Assignment Table

In this section you will find a table with an overview of the Modbus registers that are saved in the Modbus master profile for the "Moxa ioLogik E1242-T". In accordance with the XML structure, each line of the table shows the attribute values of a Modbus register assignment. The attribute values in the columns "source", "type" and "size" correspond to the standard values of the "Moxa ioLogik E1242-T".



Changed settings of the Moxa ioLogik E1242-T

If you change the settings (e.g., ip, description, etc.) of the "Moxa ioLogik E1242-T", you also have to change the attribute values of the Modbus device profile.

<Channel /> Within every channel tag, **one** source register of a slave device is assigned to the Modbus profile.
The register assignments for the individual devices can be found in the following sections.

The following channels are defined in the profile file:

Channel/name	source	destination	type	datatype	size (WORD)	swap	scale	offset	min	max	representation	unit
"DigOut_0" to "DigOut_3"	0 to 3	0 to 3	Coil	Bool	1	0	1	0	0	1	FIX0	-
"DigIn_0" to "DigIn_7"	0 to 7	0 to 7	Discrete	Bool	1	0	1	0	0	1	FIX0	-
DigIn_Value	48	48	Input	U16	1	0	1	0	0	65535	FIX0	-
"Aln_Raw_0" to "Aln_Raw_3"	512 to 515	512 to 515	Input	U16	1	0	0	0	0	65535	FIX0	-
"Aln_Float_0" to "Aln_Float_3"	520 to 526	520 to 526	Input	F32	2	0	1	0	4.0	20.0	FIX3	mA
DigOut_All	32	32	Holding	U16	1	0	1	0	0	65535	FIX0	-

6.5 Moxa ioLogik E1260-T – Assignment Table

In this section you will find a table with an overview of the Modbus registers that are saved in the Modbus master profile for the "Moxa ioLogik E1260-T". In accordance with the XML structure, each line of the table shows the attribute values of a Modbus register assignment. The attribute values in the columns "source", "type" and "size" correspond to the standard values of the "Moxa ioLogik E1260-T".



Changed settings of the Moxa ioLogik E1260-T

If you change the settings (e.g., ip, description, etc.) of the "Moxa ioLogik E1260-T", you also have to change the attribute values of the Modbus master profile.

XML tag or attribute	Explanation
<ModbusMasterProfile>	The Modbus device profile is defined within this XML structure.
</ModbusMasterProfile>	

The following channels are defined in the profile file:

Channel/name	source	destination	type	datatype	size (WORD)	swap	scale	offset	min	max	representation	unit
Clear watchdog alarm	4144	4144	Coil	Bool	1	0	1	0	0	1	FIX0	-
"Temperature_0" to "Temperature_5"	1536 to 1541	1536 to 1541	Input	U16	1	0	0.1	0	0	65535	FIX1	°C

7 User-Defined Modbus Profile

You can change the assignment of the Modbus addresses by creating a user-defined Modbus profile. In the user-defined Modbus profile, the address assignments that were defined in the SMA Modbus profile can be reassigned to other Modbus addresses. You can use the entire Modbus address range from 0 to 65535.

The user-defined Modbus profile can be called up via the gateway like an additional device and has a separate unit ID with a default setting of 100 (see Section 3.6.1 "Unit ID's", page 11).

One advantage of the user-defined Modbus profile can be that the measured values and parameters which are relevant for controlling your system can be applied to consecutive Modbus addresses and can therefore be read out or set in one block.

7.1 Structure of the XML File for the User-Defined Modbus Profile

The user-defined Modbus profile is created in the file **VirtualDevice.vdp** in addition to the SMA Modbus profile.

The basic structure of the XML file is as follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<VirtualModbusProfile>
  <Device name="Virtual device" type="virtual device" description="Merge channels of different devices to one device">
    <Mapping>
      <Channel unitid="aaa" source="bbbbbb" destination="cccc" />
      ...
    </Mapping>
  </Device>
</VirtualModbusProfile>
```

Legend for XML tags and attributes:

XML tag or attribute	Explanation
<VirtualModbusProfile>	The user-defined Modbus profile is defined within the XML master element.
<Device>	The XML element defines the name (attribute "name"), type (attribute "type") and description (attribute "description") of the Modbus profile.
<Mapping>	Source and target addresses are assigned within this XML element.
<Channel />	Within a channel tag, a Modbus address of a unit ID is redefined:
unitid="aaa"	The XML attribute specifies the unit ID of the device whose Modbus addresses are to be redefined. Possible unit IDs for individual devices are 3 to 247.
source="bbbbbb"	The XML attribute specifies a Modbus address of the device selected under "unitid" whose value is to be used as source. (Information on the assignment tables, see Section 5).
destination="cccc"	The XML attribute specifies the new Modbus address from which the value is to be retrieved (0 to 65535). Note the number of Modbus registers that are stored at the original address. The target registers must not overlap. If incomplete Modbus registers are called up later, a Modbus exception is generated. If register addresses are called up, which are not filled with values, 65535 (0xFFFF) is returned.

Modbus exceptions

 Modbus exceptions

For Modbus exceptions, see "Modbus Protocol Specification" at <https://www.modbus.org/specifications>.

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8 Integration and Troubleshooting

8.1 Notes on integration

To ensure correct control of the inverter via the Modbus interface, at least the following points should be checked and taken into account:

- ☐ The **WSpt** and **VarSpt** setpoints must be written continuously, since they are used simultaneously to monitor communication (see Section 3.11 "Content-Based Grouping of the Registers", page 16).
- ☐ The setpoints **WSpt** and **VarSpt** are to be specified in % with two decimal places. The reference value for the percentage can be read in the registers **WExlSpt.RefVal** and **VarExlSpt.RefVal**.
- ☐ Setpoints may only be sent with a limited data transfer rate. We recommend a maximum of 20 times per second (see Section 3.11 "Content-Based Grouping of the Registers", page 16).
- ☐ Parameter registers must not be written continuously. Parameter registers should only be written when changed (see Section 3.11 "Content-Based Grouping of the Registers", page 16).
- ☐ Instantaneous values cannot be read more than 10 times per second (see Section 3.11 "Content-Based Grouping of the Registers", page 16).
- ☐ The SMA registers are distributed to different unit IDs (server IDs). The unit ID used is indicated above the register tables in each case.
- ☐ Most signals are defined as double registers with the data type S32. Reading or writing to only one of the double registers is not accepted or returns invalid data.
- ☐ Registers with ENUM codes only accept the associated ENUM codes. Writing of other values is rejected as Modbus error.
- ☐ By using a customer defined Modbus profile, all registers can be defined to a common unit ID and in contiguous data blocks (see Section 7 "User-Defined Modbus Profile", page 54).

8.2 Troubleshooting

You can find information on error analysis of the SMA Modbus profile in Section 3.8 "Reading and Writing Data in the Modbus Protocol", page 13.

For troubleshooting of the SMA devices, go to Modbus address 96 and use the event numbers displayed there.

i **The event numbers of the SMA devices cannot be decrypted with the number codes in this document.**

The event numbers of the SMA devices are device-specific and cannot be decrypted with the number codes in this document.

To decrypt the event numbers of central inverters, contact the SMA Service Line (see Section 10 "Contact", page 61).

9 Note

9.1 Supported Modbus Devices from Other Manufacturers

The Modbus master profile described in this document is valid for the following Modbus-capable devices from other manufacturers:

- Moxa ioLogik E1210-T
- Moxa ioLogik E1240-T
- Moxa ioLogik E1242-T
- Moxa ioLogik E1260-T

9.2 Modbus Communication Ports

The following table shows the default settings of the supported network protocols:

Network protocol	Communication port, default setting
TCP/UDP	502



Using free communication ports

You should only use free communication ports. The following range is generally available: 49152 to 65535.

You can find more information on occupied ports in the database "Service Name and Transport Protocol Port Number Registry" at <http://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xml>.



Changing the communication port

If you change the Modbus port of the inverter, you must also change the Modbus port of a connected Modbus master system. Otherwise the inverter can no longer be accessed via the Modbus protocol.

9.3 Reaction time

The typical reaction time of the inverter is 100 ms.

The response time is the interval within which value changes in the SMA devices are available at the Modbus interface of the inverter for monitoring. Consequently, value changes in a Modbus master system (e.g., in a SCADA system) can only be displayed in a corresponding or larger interval.

9.4 Interval of the data request and number of the values

Limit of the data processing capacity

For system stability reasons, the time period between data transfers via the Modbus protocol should be at least 100 ms. 123 Modbus registers can be transferred per data block.

9.5 Number Codes of the Time Zones

The following table contains the most important time zones and their number codes in the SMA Modbus profile. If the location is known, you can determine the numerical key (code) and the time zone. In the tables in Section 5 "SMA Modbus Profile—Assignment Tables", from page 20, with specification of the time zone, this table is referenced.

City/Country	Code	Time zone
Abu Dhabi, Muscat	9503	UTC+04:00
Adelaide	9513	UTC+09:30
Alaska	9501	UTC-09:00
Amman	9542	UTC+02:00
Amsterdam, Berlin, Bern, Rome, Stockholm, Vienna	9578	UTC+01:00
Arizona	9574	UTC-07:00
Astana, Dhaka	9515	UTC+06:00
Asuncion	9594	UTC-04:00
Athens, Bucharest, Istanbul	9537	UTC+02:00
Atlantic (Canada)	9505	UTC-04:00
Auckland, Wellington	9553	UTC+12:00
Azores	9509	UTC-01:00
Baghdad	9504	UTC+03:00
Baku	9508	UTC+04:00
Bangkok, Hanoi, Jakarta	9566	UTC+07:00
Beirut	9546	UTC+02:00
Belgrade, Bratislava, Budapest, Ljubljana, Prague	9517	UTC+01:00
Bogotá, Lima, Quito	9563	UTC-05:00
Brasilia	9527	UTC-03:00
Brisbane	9525	UTC+10:00
Brussels, Copenhagen, Madrid, Paris	9560	UTC+01:00
Buenos Aires	9562	UTC-03:00
Canberra, Melbourne, Sydney	9507	UTC+10:00
Caracas	9564	UTC-04:30
Casablanca	9585	UTC+00:00
Cayenne	9593	UTC-03:00
Chennai, Kolkata, Mumbai, New Delhi	9539	UTC+05:30

Chicago, Dallas, Kansas City, Winnipeg	9583	UTC-06:00
Chihuahua, La Paz, Mazatlán	9587	UTC-07:00
Darwin	9506	UTC+09:30
Denver, Salt Lake City, Calgary	9547	UTC-07:00
Dublin, Edinburgh, Lisbon, London	9534	UTC+00:00
Yerevan	9512	UTC+04:00
Fiji, Marshall Islands	9531	UTC+12:00
Georgetown, La Paz, San Juan	9591	UTC-04:00
Greenland	9535	UTC-03:00
Guadalajara, Mexico City, Monterrey	9584	UTC-06:00
Guam, Port Moresby	9580	UTC+10:00
Harare, Pretoria	9567	UTC+02:00
Hawaii	9538	UTC-10:00
Helsinki, Kiev, Riga, Sofia, Tallinn, Vilnius	9532	UTC+02:00
Hobart	9570	UTC+10:00
Indiana (East)	9573	UTC-05:00
International Date Line (West)	9523	UTC-12:00
Irkutsk	9555	UTC+08:00
Islamabad, Karachi	9579	UTC+05:00
Yakutsk	9581	UTC+09:00
Yekaterinburg	9530	UTC+05:00
Jerusalem	9541	UTC+02:00
Kabul	9500	UTC+04:30
Cairo	9529	UTC+02:00
Cape Verde Islands	9511	UTC-01:00
Katmandu	9552	UTC+05:45
Caucasus Standard Time	9582	UTC+04:00
Krasnoyarsk	9556	UTC+07:00

Kuala Lumpur, Singapore	9544	UTC+08:00
Kuwait, Riyadh	9502	UTC+03:00
Magadan, Solomon Islands, New Caledonia	9519	UTC+11:00
Manaus	9516	UTC-04:00
Midway Islands, Samoa	9565	UTC-11:00
Minsk	9526	UTC+02:00
Mid-Atlantic	9545	UTC-02:00
Monrovia, Reykjavík	9536	UTC+00:00
Montevideo	9588	UTC-03:00
Moscow, St. Petersburg, Volgograd	9561	UTC+03:00
Nairobi	9524	UTC+03:00
Newfoundland	9554	UTC-03:30
New York, Miami, Atlanta, Detroit, Toronto	9528	UTC-05:00
Novosibirsk	9550	UTC+06:00
Nuku'alofa	9572	UTC+13:00
Osaka, Sapporo, Tokyo	9571	UTC+09:00
Pacific (U.S., Canada)	9558	UTC-08:00
Beijing, Chongqing, Hong Kong, Ürümqi	9522	UTC+08:00

Perth	9576	UTC+08:00
Petropavlovsk-Kamchatsky	9595	UTC+12:00
Port Louis	9586	UTC+04:00
Santiago	9557	UTC-04:00
Sarajevo, Skopje, Warsaw, Zagreb	9518	UTC+01:00
Saskatchewan	9510	UTC-06:00
Seoul	9543	UTC+09:00
Sri Jayawardenepura	9568	UTC+05:30
Taipei	9569	UTC+08:00
Tashkent	9589	UTC+05:00
Teheran	9540	UTC+03:30
Tbilisi	9533	UTC+04:00
Tijuana, Lower California (Mexico)	9559	UTC-08:00
Ulan Bator	9592	UTC+08:00
West-Central Africa	9577	UTC+01:00
Windhoek	9551	UTC+02:00
Vladivostok	9575	UTC+10:00
Yangon (Rangoon)	9549	UTC+06:30
Central America	9520	UTC-06:00

10 Contact

If you have technical problems with our products, please contact the SMA Service Line.

We require the following information in order to provide you with the necessary assistance:

- Modbus master software or hardware used
- Software version of your communication product
- Type of communication interface between communication product and the inverters
- Type, serial numbers, and software version of the inverters connected to your PV system

You can find your country's contact information at:



<https://go.sma.de/service>

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